

Mohamed Reda Ghalbi

AI & Data Science Engineering Student

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SUMMARY

Engineering student in Artificial Intelligence & Data Science at EMSI Casablanca with hands-on experience in predictive modelling, data pipelines and intelligent systems in banking and industrial settings. I build ML end to end, from data preparation and model training to the APIs and apps that put predictions in front of users.

EXPERIENCE

Data Science & AI Intern · Crédit du Maroc

Jul – Aug 2026

- Prepared, cleaned and transformed historical financial data for predictive modelling.
- Built a machine learning pipeline forecasting cash inflows and outflows.
- Engineered temporal lag features to improve forecasting performance.
- Contributed to the design of a decision-support system for budget tracking (Python, Pandas, Scikit-learn).

Data & AI Automation Intern · GPC (Gharb Papier et Carton)

Jul – Aug 2025

- Automated ETL pipelines (PDF, Excel, XML, REST APIs to PostgreSQL) and a Tesseract OCR pipeline for production-data extraction; real-time analytics dashboards.
- Built a desktop app automating customs certificates: PDF/OCR extraction, FIFO allocation against the PostgreSQL declarations ledger, Excel/PDF generation.

Data & Analytics Intern · FedEx

Jul – Aug 2024

- HR attendance analytics system (Flask + PostgreSQL) and automated HR KPI reporting via REST API.

PROJECTS

NutriVision AI: Food Recognition & Nutrition Estimation

Jan – Mar 2026

- Multi-task deep learning pipeline (EfficientNet-B3 + regression head) predicting calories, mass and macronutrients from a single image, with no external food database.
- MAE of 36.4 kcal on Nutrition5K, 17.3% better than Google Research's published baseline (44.0 kcal); FastAPI backend, Flutter app, LLM-generated nutrition coaching (Claude / GPT-4).

MedAI: Multi-Agent Medical Pre-Consultation

2026

- LangGraph pipeline (orchestrator, triage, doctor validation, synthesis) with FastAPI backend and Streamlit UI.

SugarSight: Diabetes Prediction Platform

Apr 2025

- End-to-end ML pipeline (Random Forest, ~81% test accuracy) with FastAPI backend and OpenAI-powered replies.

SKILLS

Languages	Python (advanced), SQL, JavaScript, HTML/CSS
ML & AI	PyTorch, Scikit-learn, EfficientNet, ResNet, multi-task regression, LangGraph, LangChain, RAG, LLM APIs (Claude, GPT-4)
Data	Pandas, NumPy, ETL pipelines, OCR (Tesseract), Matplotlib, Seaborn, PostgreSQL, MongoDB
Engineering	FastAPI, Flask, REST APIs, React, Flutter, Git, Linux, Google Colab

CERTIFICATIONS

Deep Learning Specialization, DeepLearning.AI, 2026 (5 courses: Neural Networks and Deep Learning · Improving Deep Neural Networks · Structuring ML Projects · CNNs · Sequence Models)
CS50x, HarvardX, 2024 · **Agile Project Management**, Google

EDUCATION

Engineering Degree: Computer Science, AI & Data Science · EMSI Casablanca

2022 – present

Baccalaureate in Physical Sciences

2022

LANGUAGES & ACTIVITIES

Languages: Arabic (native) · French (fluent) · English (fluent) · German (beginner)

Judo: 2nd place, Coupe du Trône (national), 2024 · Judo referee

Leadership: Organizer, 11th EMSI Careers Forum (2025) · Class delegate, EMSI